

Undergraduate Research Proposal — Why the Neutrino Mass Bound Moves in Extended Cosmologies

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Summary

This project takes one result from [Giarè, Lee & Di Valentino \(2026\)](#) and asks Loverde's question of it. Their Table 4 shows the 95% upper bound on the total neutrino mass rising from about 0.061 eV in Λ CDM to 0.113–0.123 eV once dark energy evolves, and to 0.171–0.184 eV when every sector is opened. They report how far the bound moves, but not which physical channel moves it.

[Loverde & Weiner \(2024\)](#) supply a way to answer that: split a neutrino mass constraint into what it gets from geometry and what it gets from the growth of structure. The project applies that split to three rungs of the Intertwined model ladder and asks which channel relaxes.

The Intertwined paper already points toward Loverde's work. It traces the Planck–DESI disagreement largely to a mismatch in the matter density parameter, citing her paper among others [222], and it finds that matter density shifts upward in the evolving-dark-energy models while the clustering amplitude barely changes. Connecting those two observations is the whole project.

Background

Cosmology currently bounds the neutrino mass so tightly that it presses against the minimum oscillation experiments allow, and the Intertwined paper shows how fragile that squeeze is. Oscillation data require a total mass of at least about 0.059 eV for normal ordering and 0.10 eV for inverted ordering.

What the Intertwined paper found

They fit CMB (Planck, ACT DR6, SPT-3G, lensing, BICEP/Keck), DESI DR2 BAO, and one of two supernova samples (Pantheon-Plus or DES-Dovekie). They then open one sector of the model at a time. The neutrino results that matter here, all at 95% confidence with

CMB+DESI+DD:

Model	Upper bound on total mass	Weight below the normal-ordering floor
LambdaCDM + mass	0.061 eV	about 94%
LambdaCDM + mass + N_{eff} + curvature + inflation parameters	0.122 eV	about 68%
$w_0 w_a$ CDM + mass	0.123 eV	about 64%
Everything open, including evolving dark energy	0.184 eV	about 44%

Three of their conclusions set up this project:

- The relaxation comes from added late-time background freedom. They name evolving dark energy and spatial curvature as the two levers, which act on distances rather than on clustering.
- The matter density moves between model branches while the clustering amplitude σ_8 stays within about 1%. That is a geometry-shaped fingerprint, but they do not test it as one.
- The preference for normal ordering and the tension with oscillation data both collapse as the model is extended, so neither is a robust result of cosmology alone.

What Loverde's work adds

Loverde & Weiner showed that geometry is at least as important as structure suppression in neutrino mass inference. Within LambdaCDM, the CMB alone forces the matter fraction to rise steeply with neutrino mass. Lensing observables depend on the matter fraction, not the matter density. And DESI's tight bound comes mostly from geometric incompatibility with massive neutrinos.

If that is right, then evolving dark energy and curvature should relax the geometric channel and leave the growth channel nearly untouched. Nobody has checked this directly across a model ladder like Intertwined's. That gap is the project.

Research question

When extending the cosmological model roughly doubles the neutrino mass bound, is it the geometric constraint or the growth constraint that relaxes?

Three rungs of the Intertwined ladder are enough to answer it: Λ CDM + mass, w_0 w Λ CDM + mass, and Λ CDM + curvature + mass. Three hypotheses:

1. **H1 (geometry does the work).** Going from Λ CDM to w_0 w Λ CDM, the geometry-only bound relaxes by most of the factor of two seen in the full bound, while the growth-only bound barely moves. This follows if Loverde & Weiner are right and matches the stable σ_8 in the Intertwined paper.
2. **H2 (curvature takes the same route).** Opening curvature instead of dark energy relaxes the bound through the same geometric channel, since both change distances rather than clustering.
3. **H3 (supernova sample matters in the split).** Intertwined finds the two supernova samples give nearly the same full bound. Loverde & Weiner found supernovae pull the geometric constraint in the opposite direction from BAO. Those can both be true only if the difference cancels in the combination, and the split would show it.

H1 is the core deliverable. H2 and H3 are extensions, and one of them is enough.

Methods

The plan borrows the Intertwined paper’s model ladder and Loverde & Weiner’s decomposition, but not the Intertwined paper’s full likelihood. Their pipeline is CAMB plus Cobaya with full Planck, ACT DR6 and SPT-3G likelihoods, run to convergence for 18 models and two dataset combinations. That is cluster-scale work and not a realistic starting point for an undergraduate.

Step	What it involves	Tooling
1. Warm-up	Rebuild the Intertwined ordering diagnostics (Bayes factor, tail probability below the oscillation floor, overlap coefficient) on a public chain. These are post-processing, a few dozen lines	NumPy, GetDist

Step	What it involves	Tooling
2. Lighter reproduction	Run Λ CDM + mass and w_0 wCDM + mass with a reduced CMB likelihood, DESI DR2 BAO and Pantheon-Plus; check the bound roughly doubles	Cobaya, CAMB
3. Build the split	Separate the constraint into a geometry part and a growth part following Loverde & Weiner	Custom Python
4. Test H1	Compare how each part moves between the two models	Same chains
5. Extend	Curvature rung (H2) or supernova swap (H3)	Same machinery

The validation target is qualitative agreement with the Intertwined Table 4. Exact numbers are not expected with a lighter CMB likelihood, but the ordering of bounds and the direction of the matter-density shift should reproduce. If they do not, the reduced likelihood is losing the effect and the project needs to change before it goes further.

What I would need from the group: whether a reduced CMB likelihood is acceptable for this question, and whether anyone already has chains for these models. The Intertwined authors have posted some analysis tools publicly, though I did not find the chains themselves.

Timeline

Two quarters, with a checkpoint at each stage that either passes or triggers a conversation.

Weeks	Work	Checkpoint
1-3	Read the Intertwined neutrino section and Loverde & Weiner; rebuild the ordering diagnostics on a public chain	My tail probability and overlap numbers behave the way their Section 5.3 describes
4-6	Run Λ CDM + mass and w_0 wCDM + mass with the lighter likelihood	The bound roughly doubles between the two, as in their Table 4

Weeks	Work	Checkpoint
7-10	Implement the geometry/growth split	I can state each channel's bound for both models
11-14	Test H1 and make the main figure	One figure showing which channel relaxes
15-20	One extension (H2 or H3), write up	Draft, poster, or a section of a paper

Weeks 4 to 6 are the real gate. If I cannot recover a published number, the rest of the plan is not worth starting, and that failure is cheap and fast to discover.

Deliverables

- A documented Python repository that takes a chain or a model choice and returns geometry-only and growth-only neutrino mass constraints, plus the ordering diagnostics.
- One main figure: how the geometric and growth bounds move from LambdaCDM to w_0wa CDM.
- A written report suitable for a Mary Gates application, a departmental poster, or a methods section.

A null result is still a result. If the growth channel turns out to relax as much as the geometric one, that contradicts the simplest reading of Loverde & Weiner in extended models, and she would want to know. The failure that would actually sink the project is not reproducing the direction of the Table 4 trend, which is why that check sits at week six.

Key readings

Bracketed numbers are entries in the Intertwined paper's reference list. Read in this order.

Paper	arXiv	Why
Giarè, Lee & Di Valentino, Intertwined Constraints in Extended Cosmologies	2607.01226	The anchor. Sections 2 and 5 first, then Table 7 and Figure 12 on the matter density

Paper	arXiv	Why
Loverde & Weiner, Massive neutrinos and cosmic composition [222]	2410.00090	The method, and the connection to Loverde
Lesgourgues & Pastor, Massive neutrinos and cosmology [217]	astro-ph/0603494	Standard background on neutrino physics in cosmology
DESI DR2 BAO cosmological constraints [40]	2503.14738	The geometric dataset at the centre of the tension
Elbers et al., Negative neutrino masses as a mirage of dark energy [226]	2407.10965	The dark-energy explanation, closest to H1
Lynch & Knox, What's the matter with the neutrino mass sum? [231]	2503.14470	Another matter-density reading of the same tension
Green & Meyers, Cosmological preference for a negative neutrino mass [225]	2407.07878	States the negative-mass puzzle
Lewis, GetDist [245]	1910.13970	Practical, for the plots

The Intertwined paper is a 2026 preprint, now in its second version, and has not been through peer review. Treat its numbers as provisional and recheck them against the latest arXiv version before citing them.

How to pitch it

Don't send the proposal first. Send a short email and attach it only if she asks.

The pitch is stronger now because you are bringing her something she may not have read closely. A working shape, under 150 words:

1. The observation: the Intertwined paper finds the neutrino mass bound roughly doubles once dark energy evolves, while the matter density shifts and σ_8 barely moves.
2. The question: whether that relaxation is the geometric channel from her 2024 paper with Weiner, and whether anyone has split it.
3. What you can do: Python, and whether you have used Cobaya, CAMB or MCMC before.

4. The ask: whether the group has room for an undergraduate, or whether you could sit in on a group meeting.

Before sending, check the group's recent seminar talks. Her students have spoken on the sound speed of nonlinear structure and on how the CMB constrains dark radiation. If a current student is already doing this split, you want to find that out first and offer to help rather than propose it cold.

Two cautions. The reduced likelihood is the part a cosmologist will question first, so say up front that you know it is an approximation and want her view. And a proposal gets a meeting, not a letter. What gets a letter is two quarters of showing up and doing the work, so hold the plan loosely and let her reshape it.